

# Mark Jarzynski

Graphics Engineer | Ph.D Candidate

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[markjarzynski.github.io](https://markjarzynski.github.io)

[mastodon.gamedev.place/@markjarzynski](https://mastodon.gamedev.place/@markjarzynski)

## Games

### **Ashes of the Singularity II**

TBD

Large-scale RTS game. Nitrous Engine 3. Developed and maintained graphics features and shaders. Created the new terrain shader with stochastic hex tiling. Added area lights (LTC). Optimized shader code and fixed bugs. Implemented many of the orbitals in the game.

### **Sins of a Solar Empire II: Paths to Power**

2025

4X space strategy game. Iron Engine 3. Fixed many graphics bugs in the base game. GPU and CPU performance optimization. Implemented lua scripting, and scripted many of the scenarios introduced in Paths to Power DLC.

### **Ara History Untold**

2024

4X grand strategy game. Nitrous Engine 2. Developed and maintained many of the graphics features. Responsible for much of the procedural generation, such as the City Renderer.

## Work Experience

### **Graphics Engineer, Oxide Games**

2021 - present

- Worked in a custom game engine, Nitrous, and maintained its Decoupled Shading Engine.
- Developed and maintained Graphics features and shaders. Terrain, water, point lights, area lights, effects, particles, trails, GGX, OrenNayar, etc.
- Worked closely with artist, tech artist, and production to achieve the desired look and to make sure our tools were meeting their needs.
- Wrote highly parallelized and high performant SSE code in C++.

### **Technical Director, UMBC, Imaging Research Center**

2011 - 2021

- Supervised, led, and educated teams of students and staff on research projects.
- Lead Developer on research projects, often involve the following:
  - VR, AR, MR
  - Unreal Engine / Unity
  - 3D Scanning / Photogrammetry
  - Motion Tracking / Motion Capture
  - Data Visualization and Procedural Generation

- Parallel Computing
- Maintained department's infrastructure, servers, desktops.

## Systems Administrator & Programmer

2011 - 2014

University of Maryland Baltimore County, Imaging Research Center

- Developed and maintained mission critical software to meet research needs.
- Worked with students to ensure researcher needs were being met.
- Maintained department's infrastructure, servers, desktops.

## Education

|       |                                                                                                                                                                                                      |         |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Ph.D. | Computer Science; University of Maryland, Baltimore County<br>Dissertation: Hashing Out Performance, Evaluating and Optimizing Hash Functions for GPU Rendering<br>Admitted to Candidacy: July 2024. | present |
| B.S.  | Computer Science; University of Maryland, Baltimore County                                                                                                                                           | 2011    |

## Technical Skills

|                     |                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------|
| Languages:          | C, C++, C#, Python, Java, Php, HTML, CSS, Javascript, HLSL, GLSL                                                    |
| Graphics APIs:      | DirectX 11/12, Vulkan, OpenGL                                                                                       |
| Graphics Debuggers: | PIX, Nvidia Nsight Aftermath, Radeon Developer Tool Suite                                                           |
| Game Engines:       | Nitrous, Iron Engine, Unreal Engine, Unity                                                                          |
| Operating Systems:  | Linux, macOS, Windows                                                                                               |
| Databases:          | MariaDB, MySQL, MSSQL, Postgres, Oracle                                                                             |
| Software:           | Visual Studio, XCode, CMake, Makefiles, GCC, LLVM, Perforce, Git, Subversion, CVS, Mercurial, Apache, NGINX, Docker |

## Publications

Dan Baker and Mark Jarzynski, Generalized Decoupled and Object Space Shading System, *Eurographics Symposium on Rendering (ESGR)*, 2022  
DOI: <https://doi.org/10.2312/sr.20221163>

Mark Jarzynski and Marc Olano, Hash Functions for GPU Rendering, *Journal of Computer Graphics Techniques (JCGT)*, vol. 9, no. 3, 21-38, 2020  
Available online <http://jcgt.org/published/0009/03/02/>

## Awards

2006 Eagle Scout. Boy Scouts of America. Troop 447.